

Project 2 Report: Employee Attrition

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Abstract This project develops a blank.

Introduction

The objective of this project is to develop a predictive model capable of determining which employees are most likely to leave. This can allow the business to focus their retention efforts on high-risk employees.

The business decision addressed by this analysis is:

Which employees are likely to leave?

The resulting model can be used to identify employees that appear likely to leave.

Business Objective

The primary objective is to accurately determine which employees are likely to quit before they do.

A practical utility function for this problem is:

$$U = \begin{cases} 0.60 \times 40000 - 8000 = 16000 & \text{if a retention intervention is provided and prevents attrition} \\ -8000 & \text{if a retention intervention is provided unnecessarily} \\ -40000 & \text{if no intervention is provided and the employee leaves} \\ 0 & \text{if no intervention is provided and the employee stays} \end{cases}$$

assumptions:

- Cost of employee turnover: \$40,000
- Cost of retention intervention: \$8,000
- Retention intervention success rate: 60%

Plan

The analysis follows the regression modeling workflow:

1. Define the business problem.
2. Acquire and clean employee data.
3. Explore relationships between candidate predictors and attrition.
4. Build an interpretable regression model.
5. Evaluate assumptions and model fit.
6. Generate predictions and business recommendations.

The ideal dataset would contain:

- Age
- Attrition
- Salary
- Years at the company
- Overtime
- Job role
- Job satisfaction score

Due to the data being fake generated data, it was more than ideal. It included everything we thought of and much more.

Build

Data Source

The dataset was obtained from Kaggle at <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset> and contains a fictional dataset created by IBM data scientists.

Variables initially available included 35 different point including notably:

- Age
- Attrition
- Department
- Gender
- Job Role
- Job Satisfaction
- Martial Status
- Monthly Income
- Number of Companies Worked
- Overtime
- Years Since Last Promotion
- Year at Company
- Years in Current Role

Data Cleaning

The following preprocessing steps were performed:

- Converted Attrition to a binary outcome where Yes = 1

```

import os
import numpy as np
import pandas as pd
from patsy import dmatrices
import polars as pl
import seaborn.objects as so
import seaborn as sns
import matplotlib.pyplot as plt
import statsmodels.api as sm
import statsmodels.formula.api as smf
from sklearn.model_selection import train_test_split
from statsmodels.stats.outliers_influence import variance_inflation_factor

pl.Config.set_tbl_rows(-1)
pl.Config.set_tbl_cols(-1)
pl.Config.set_fmt_str_lengths(100)

employee_data = (pl.read_csv(os.path.join('data', 'HR-Employee-Attrition.csv')))
# Make the outcome binary where 'Yes' = 1
.with_columns(
    pl.when(pl.col('Attrition') == 'Yes').then(1)
    .when(pl.col('Attrition') == 'No').then(0)
    .alias('Left')
)
.select(pl.exclude(['EmployeeNumber', 'DailyRate', 'MonthlyIncome',
'EmployeeCount', 'Over18', 'PerformanceRating']))
)

# Drop missing values in key columns
# df = raw_data.drop_nulls(subset=['make', 'body', 'color', 'condition', 'year',
'saledate', 'sellingprice', 'state', 'transmission'])

# Print unique values and their counts
# for col in employee_data.columns:
#     print(f"\n{'='*60}")
#     print(f"Column: {col}")
#     print(f"{'='*60}")

#     print(
#         employee_data
#         .group_by(col)
#         .len()
#         .sort("len", descending=True)

```

```

# )

predictors = [
    "Age",
    "C(BusinessTravel)",
    # "C(Department)", # job role perfectly maps to certain departments except
manager
    "DistanceFromHome",
    "C(EducationField)", # This seems to be redundant.
    "C(Education)",
    "EnvironmentSatisfaction",
    "C(Gender)",
    "HourlyRate", # Hourly rate seems to group employees together better than other
similar rate columns that tend to only match one or few employees.
    "JobInvolvement",
    "C(JobLevel)",
    "C(JobRole)",
    "JobSatisfaction",
    "C(MaritalStatus)",
    "MonthlyRate", # may be more linear and likely redundant with hourly rate
    "NumCompaniesWorked",
    "C(OverTime)",
    "PercentSalaryHike",
    "RelationshipSatisfaction",
    "TotalWorkingYears", # may correlate strongly with Age
    "TrainingTimesLastYear",
    "WorkLifeBalance",
    "YearsAtCompany",
    "YearsInCurrentRole",
    "YearsSinceLastPromotion",
    "YearsWithCurrManager"
]

```

Explore

Exploratory Data Analysis

Initial visualizations focused on the relationships between attrition and major numerical predictors.

Overtime vs Attrition

```

overtime_rate = (
    employee_data
    .group_by("OverTime")

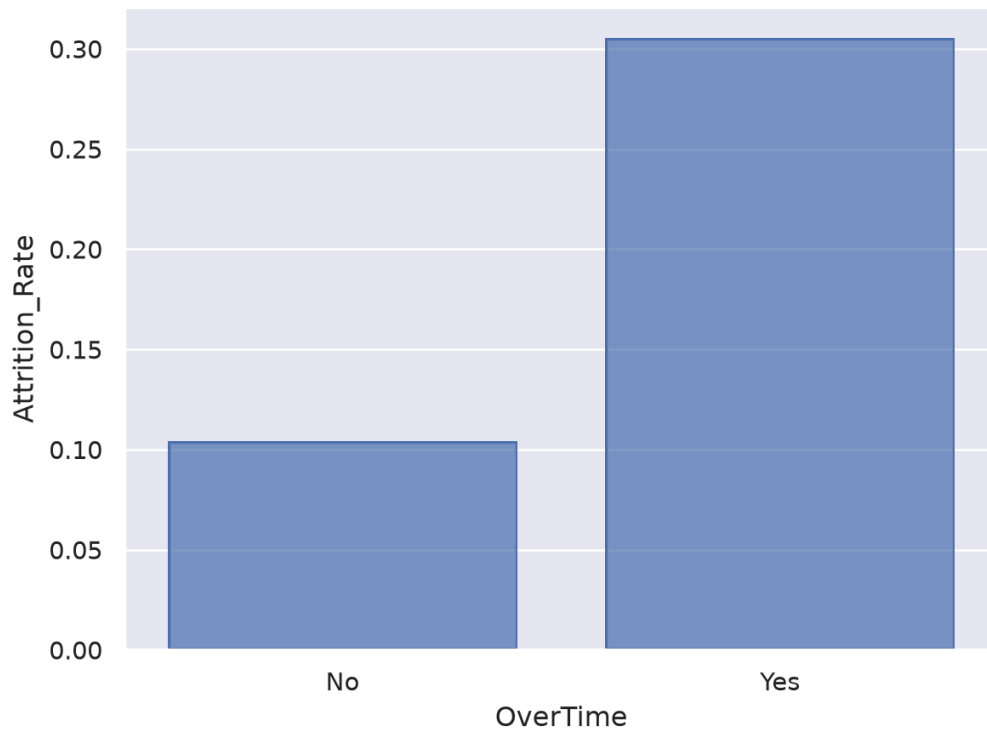
```

```

        .agg(Attrition_Rate=pl.col("Left").mean())
    )

so.Plot(overtime_rate.to_pandas(),
        x="OverTime",
        y="Attrition_Rate") \
    .add(so.Bar())

```



This looks like employees that work overtime are more likely to leave.

Satisfaction vs Attrition

```

(
    so.Plot(employee_data.to_pandas(),
            x="JobSatisfaction",
            y="Left")
        .add(so.Dot(alpha=.20))
        .add(so.Line(), so.PolyFit(order=1))
    ).show()

(
    so.Plot(employee_data.to_pandas(),

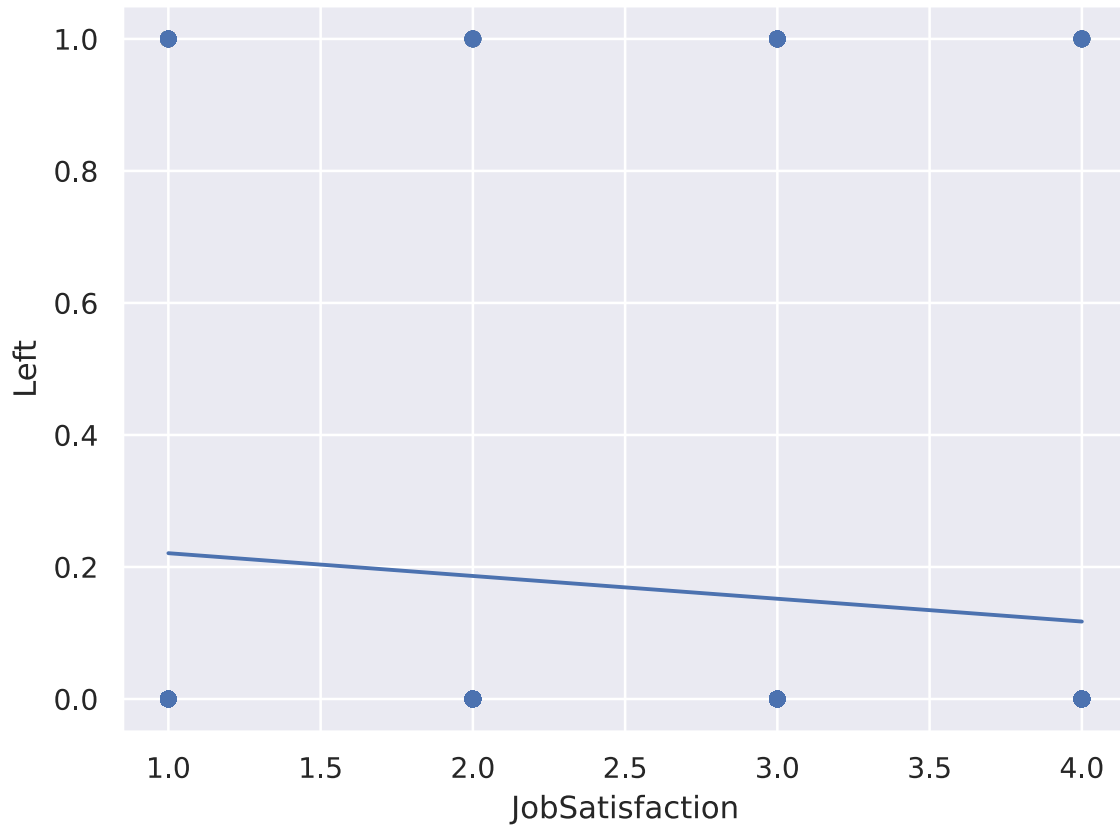
```

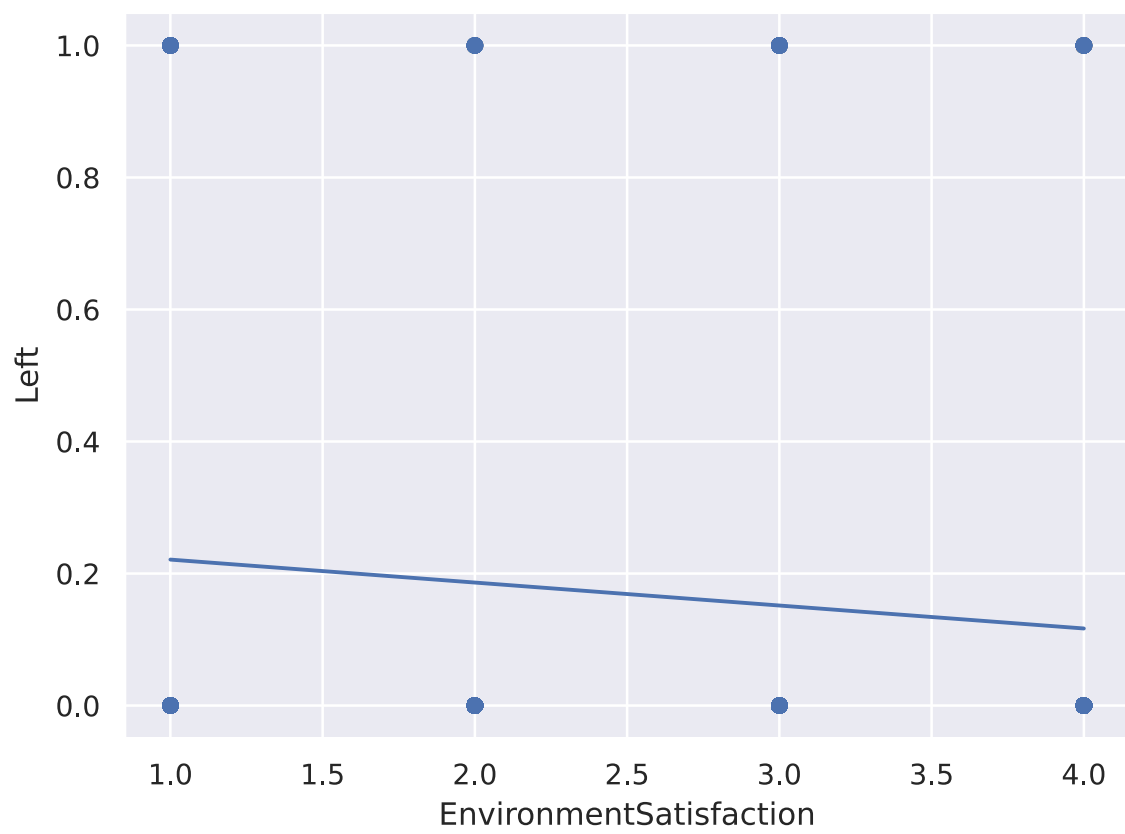
```

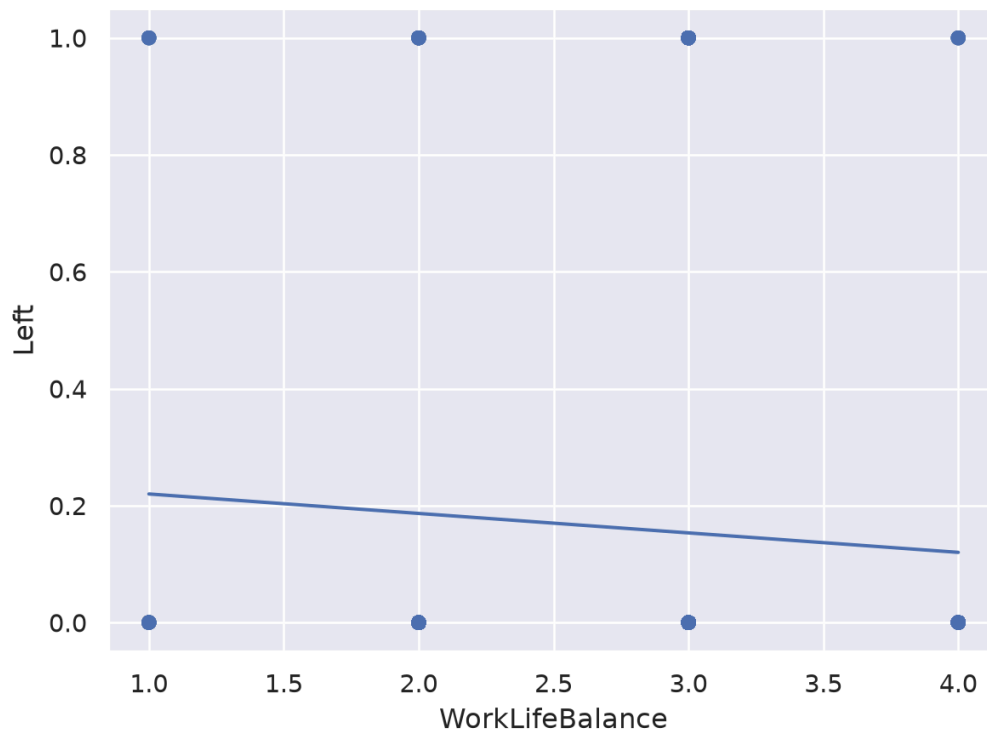
        x="EnvironmentSatisfaction",
        y="Left")
    .add(so.Dot(alpha=.20))
    .add(so.Line(), so.PolyFit(order=1))
).show()

(
    so.Plot(employee_data.to_pandas(),
        x="WorkLifeBalance",
        y="Left")
    .add(so.Dot(alpha=.20))
    .add(so.Line(), so.PolyFit(order=1))
)

```



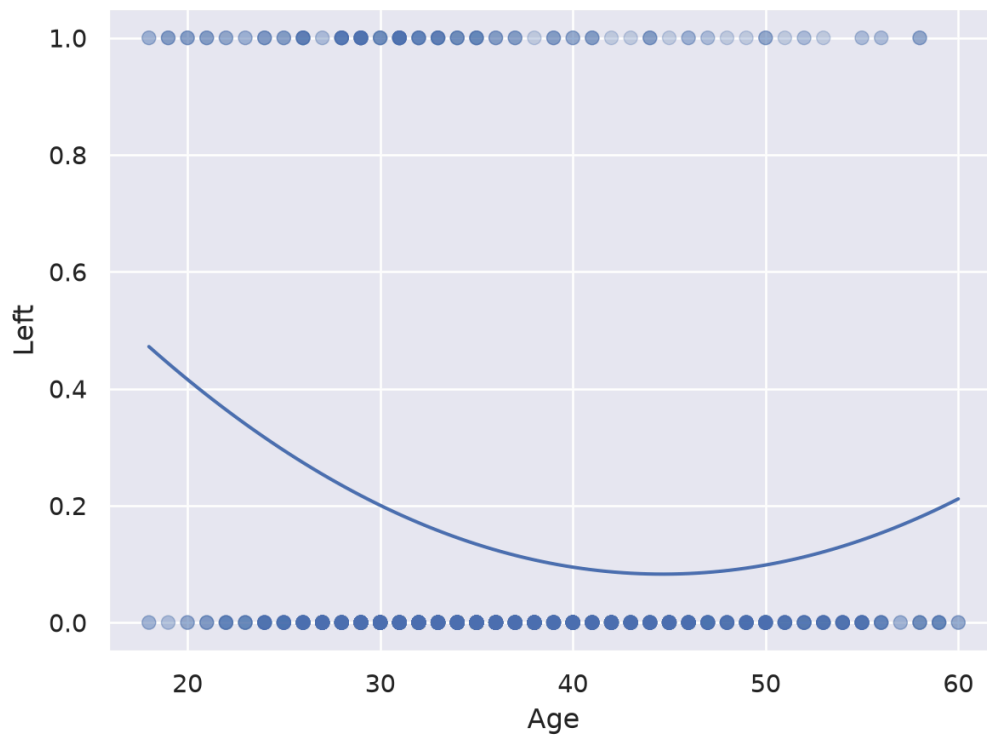




These graphs show what we would expect, that as employees report being more satisfied or having better work/life balance, they are less likely to leave.

Age vs Attrition

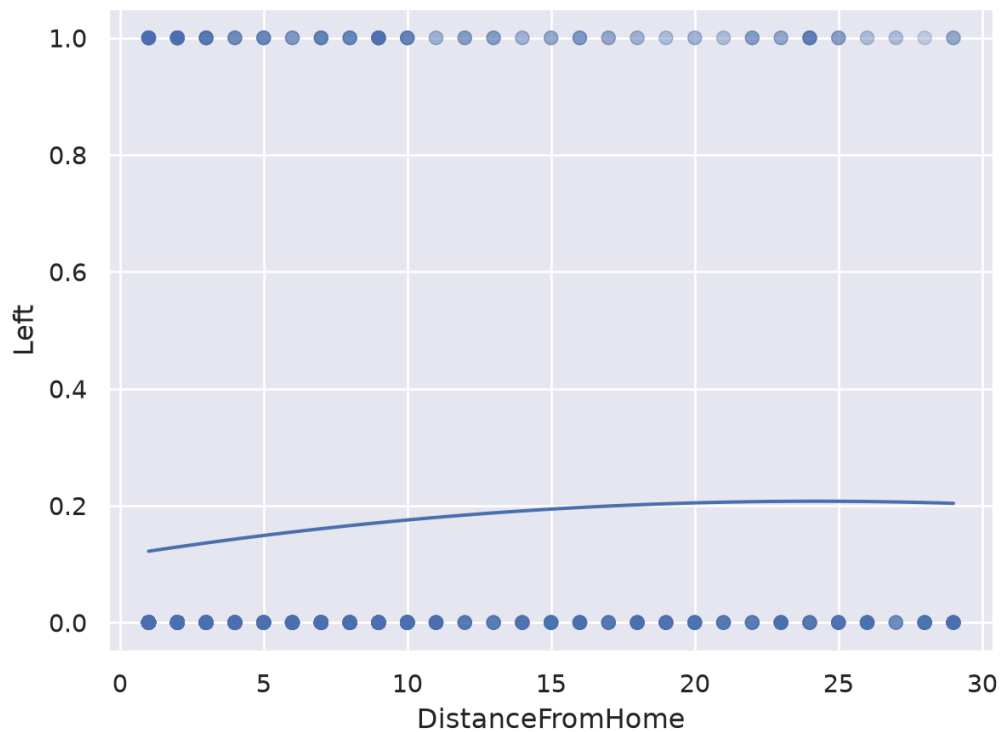
```
(  
    so.Plot(employee_data.to_pandas(),  
            x="Age",  
            y="Left")  
    .add(so.Dot(alpha=.15))  
    .add(so.Line(), so.PolyFit(order=2))  
)
```

We can see that younger employees are generally more likely to leave, but this starts to reverse around age 45. Retirement is likely a factor, but also employees are likely changing jobs for more senior roles.

Commute vs Attrition

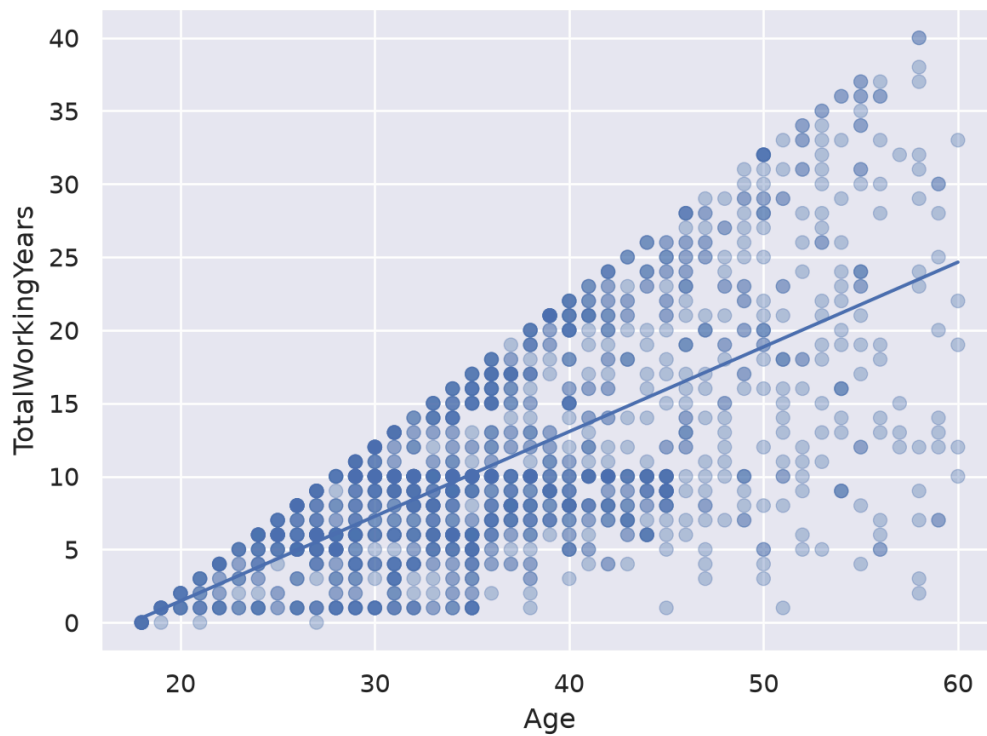
```
(  
    so.Plot(employee_data.to_pandas(),  
            x="DistanceFromHome",  
            y="Left")  
    .add(so.Dot(alpha=.15))  
    .add(so.Line(), so.PolyFit(order=2))  
)
```



There is a fairly weak correlation between how far someone lives from work and if they are likely to leave. It is interesting that the trend appears to level off so it seems like it's only a factor within a certain range.

Age vs Total Working Years

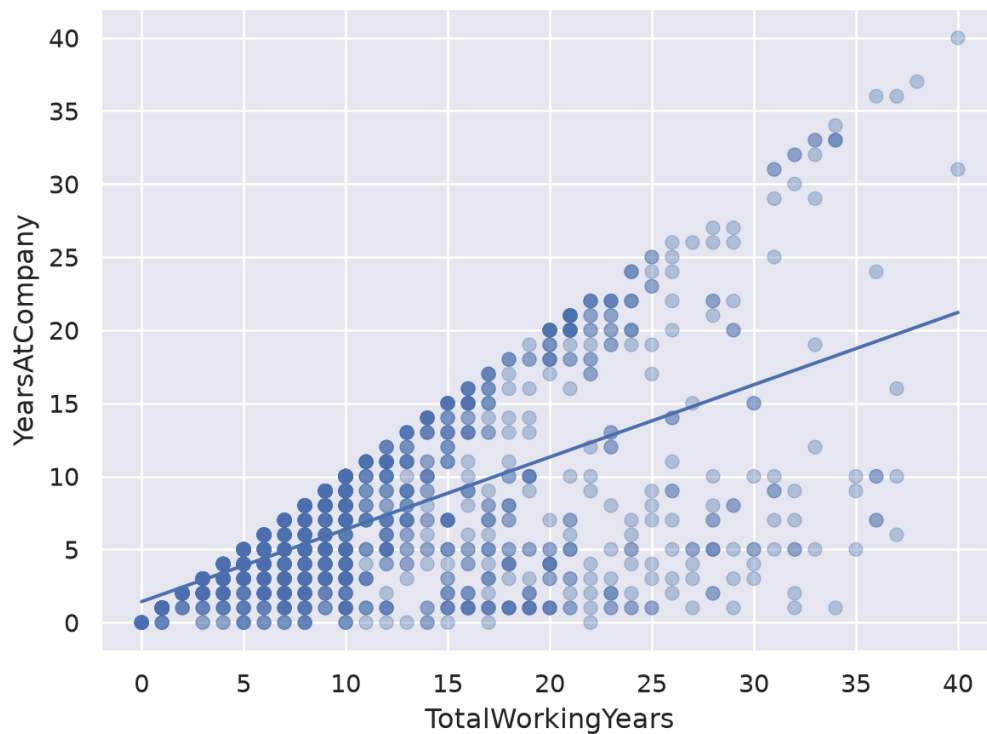
```
(
    so.Plot(employee_data.to_pandas(),
            x="Age",
            y="TotalWorkingYears")
    .add(so.Dot(alpha=.35))
    .add(so.Line(), so.PolyFit(order=1))
)
```



We thought there would be a strong relationship in the data that could cause issues between employee age and total working years. It looks like there is a relationship, but the correlation is not nearly as strong as we thought.

Years at Company vs Total Working Years

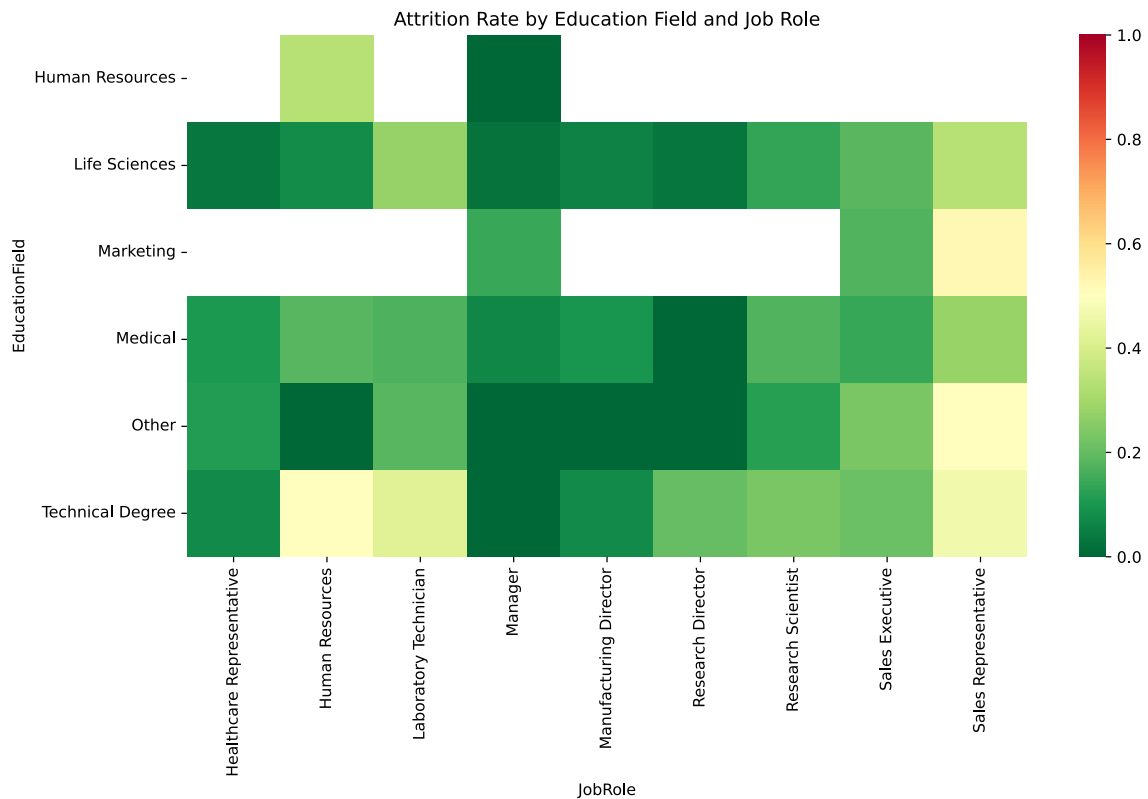
```
(  
    so.Plot(employee_data.to_pandas(),  
            x="TotalWorkingYears",  
            y="YearsAtCompany")  
    .add(so.Dot(alpha=.35))  
    .add(so.Line(), so.PolyFit(order=1))  
)
```



It looks like there is a relationship between years at the company and total working years, but not a very strong one.

Education Field vs Job Role

```
rate = (
    employee_data
    .group_by(["EducationField", "JobRole"])
    .agg(attrition_rate=pl.col("Left").mean())
    .to_pandas()
)
pivot_rate = rate.pivot(
    index="EducationField",
    columns="JobRole",
    values="attrition_rate"
)
plt.figure(figsize=(12,6))
sns.heatmap(pivot_rate, cmap="RdYlGn_r", vmin=0, vmax=1)
plt.title("Attrition Rate by Education Field and Job Role")
plt.show()
```

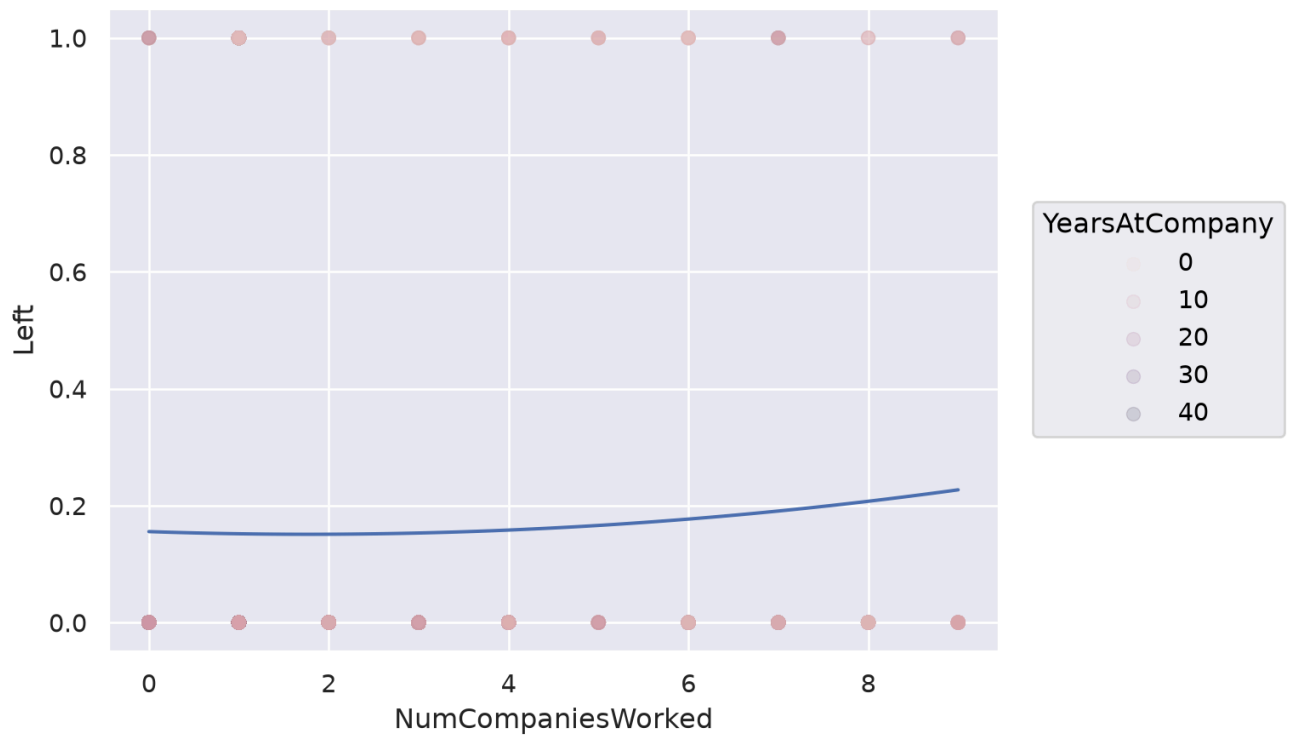


The heatmap reveals a strong structural relationship between education field and job role, indicating that employees are not randomly distributed across roles. Instead, specific education backgrounds concentrate within particular job families. This dependency explains multicollinearity we later observed in the initial logistic regression model.

Attrition varies more meaningfully across job roles than across education fields, suggesting that occupational role is a stronger predictor of turnover than educational background.

Number of Companies and Attrition

```
(
  so.Plot(employee_data.to_pandas(),
    x="NumCompaniesWorked",
    y="Left")
  .add(so.Dot(alpha=.15), color="YearsAtCompany")
  .add(so.Line(), so.PolyFit(order=2))
)
```



Key Findings

- Found this.
- Found that.

Reconcile

Sampling Strategy

A stratified sampling approach was used to evenly split test and training data with even attrition and non-attrition employees.

Assumptions

Model Refinements

Several iterations of the model were evaluated.

Adjustments included:

- abcd
- GDs.
- Evaluating Variance Inflation Factors (VIFs).

```
pb_train = employee_data.to_pandas()

train_df, test_df = train_test_split(
```

```

    pb_train,
    test_size=0.30,
    random_state=42,
    stratify=pb_train["Left"]
)

formula = "Left ~ " + " + ".join(predictors)

fit_01 = smf.logit(
    formula,
    data=train_df
).fit()

print(fit_01.summary())

# Create the design matrix used by the model
y, X = dmatrices(formula, data=pb_train, return_type="dataframe")

# Calculate VIF
vif = pd.DataFrame({
    "Variable": X.columns,
    "VIF": [variance_inflation_factor(X.values, i) for i in range(X.shape[1])]
})

print(vif.sort_values("VIF", ascending=False))

```

```

Optimization terminated successfully.
    Current function value: 0.271364
    Iterations 9

```

```

                        Logit Regression Results
=====
Dep. Variable:          Left    No. Observations:          1029
Model:                  Logit    Df Residuals:              984
Method:                  MLE     Df Model:                 44
Date:                   Sat, 04 Jul 2026    Pseudo R-squ.:         0.3859
Time:                   21:36:07    Log-Likelihood:         -279.23
converged:              True      LL-Null:               -454.67
Covariance Type:        nonrobust    LLR p-value:            1.916e-49
=====

```

	coef	std err	z	P>
z	[0.025	0.975]		

```

-----

```

Intercept	4.1633	1.769	2.354	0.019
0.697 7.630				
C(BusinessTravel)[T.Travel_Frequently]	2.0156	0.514	3.919	0.000
1.008 3.024				
C(BusinessTravel)[T.Travel_Rarely]	1.0539	0.467	2.254	0.024
0.138 1.970				
C(EducationField)[T.Life Sciences]	-1.7028	0.933	-1.825	0.068
-3.532 0.126				
C(EducationField)[T.Marketing]	-1.4330	0.980	-1.463	0.144
-3.353 0.487				
C(EducationField)[T.Medical]	-1.8102	0.936	-1.933	0.053
-3.646 0.025				
C(EducationField)[T.Other]	-2.3162	1.057	-2.191	0.028
-4.388 -0.245				
C(EducationField)[T.Technical Degree]	-1.2318	0.952	-1.294	0.196
-3.098 0.634				
C(Education)[T.2]	0.0728	0.406	0.179	0.858
-0.722 0.868				
C(Education)[T.3]	0.1837	0.347	0.529	0.597
-0.497 0.865				
C(Education)[T.4]	0.0287	0.383	0.075	0.940
-0.721 0.778				
C(Education)[T.5]	0.1043	0.750	0.139	0.889
-1.366 1.575				
C(Gender)[T.Male]	0.3303	0.234	1.411	0.158
-0.128 0.789				
C(JobLevel)[T.2]	-1.4576	0.526	-2.770	0.006
-2.489 -0.426				
C(JobLevel)[T.3]	-0.0543	0.680	-0.080	0.936
-1.387 1.278				
C(JobLevel)[T.4]	-0.6839	1.083	-0.632	0.528
-2.806 1.438				
C(JobLevel)[T.5]	1.0270	1.409	0.729	0.466
-1.734 3.788				
C(JobRole)[T.Human Resources]	0.9110	0.942	0.968	0.333
-0.934 2.756				
C(JobRole)[T.Laboratory Technician]	1.2668	0.722	1.754	0.079
-0.149 2.682				
C(JobRole)[T.Manager]	-0.1943	0.998	-0.195	0.846
-2.151 1.762				
C(JobRole)[T.Manufacturing Director]	0.5552	0.721	0.770	0.441
-0.858 1.968				
C(JobRole)[T.Research Director]	-2.0340	1.390	-1.464	0.143

-4.758	0.690				
C(JobRole)[T.Research Scientist]		-0.1597	0.762	-0.210	0.834
-1.654	1.334				
C(JobRole)[T.Sales Executive]		1.6846	0.606	2.782	0.005
0.498	2.872				
C(JobRole)[T.Sales Representative]		1.7233	0.815	2.115	0.034
0.126	3.321				
C(MaritalStatus)[T.Married]		0.4107	0.321	1.279	0.201
-0.219	1.040				
C(MaritalStatus)[T.Single]		1.6026	0.336	4.769	0.000
0.944	2.261				
C(OverTime)[T.Yes]		2.0727	0.251	8.273	0.000
1.582	2.564				
Age		-0.0420	0.018	-2.390	0.017
-0.076	-0.008				
DistanceFromHome		0.0502	0.014	3.714	0.000
0.024	0.077				
EnvironmentSatisfaction		-0.4987	0.106	-4.693	0.000
-0.707	-0.290				
HourlyRate		-0.0025	0.006	-0.446	0.656
-0.013	0.008				
JobInvolvement		-0.5621	0.150	-3.747	0.000
-0.856	-0.268				
JobSatisfaction		-0.3862	0.101	-3.838	0.000
-0.583	-0.189				
MonthlyRate		2.351e-05	1.56e-05	1.504	0.133
-7.14e-06	5.42e-05				
NumCompaniesWorked		0.2196	0.049	4.499	0.000
0.124	0.315				
PercentSalaryHike		-0.0171	0.031	-0.547	0.584
-0.078	0.044				
RelationshipSatisfaction		-0.3647	0.102	-3.560	0.000
-0.566	-0.164				
TotalWorkingYears		-0.0687	0.036	-1.903	0.057
-0.140	0.002				
TrainingTimesLastYear		-0.1471	0.091	-1.618	0.106
-0.325	0.031				
WorkLifeBalance		-0.4242	0.158	-2.690	0.007
-0.733	-0.115				
YearsAtCompany		0.0197	0.052	0.380	0.704
-0.082	0.121				
YearsInCurrentRole		-0.1245	0.060	-2.060	0.039
-0.243	-0.006				

YearsSinceLastPromotion	0.1914	0.053	3.594	0.000
0.087	0.296			
YearsWithCurrManager	-0.0920	0.063	-1.466	0.143
-0.215	0.031			

	Variable	VIF
0	Intercept	230.084646
3	C(EducationField)[T.Life Sciences]	20.401900
5	C(EducationField)[T.Medical]	18.316684
4	C(EducationField)[T.Marketing]	9.216272
7	C(EducationField)[T.Technical Degree]	7.600715
6	C(EducationField)[T.Other]	5.234651
38	TotalWorkingYears	5.098665
41	YearsAtCompany	4.781034
15	C(JobLevel)[T.4]	4.736300
16	C(JobLevel)[T.5]	4.292659
22	C(JobRole)[T.Research Scientist]	4.251536
14	C(JobLevel)[T.3]	3.908152
18	C(JobRole)[T.Laboratory Technician]	3.898645
13	C(JobLevel)[T.2]	3.749742
23	C(JobRole)[T.Sales Executive]	3.172551
19	C(JobRole)[T.Manager]	2.999902
44	YearsWithCurrManager	2.902772
42	YearsInCurrentRole	2.850916
9	C(Education)[T.3]	2.799689
10	C(Education)[T.4]	2.669351
2	C(BusinessTravel)[T.Travel_Rarely]	2.384178
1	C(BusinessTravel)[T.Travel_Frequently]	2.371898
24	C(JobRole)[T.Sales Representative]	2.317760
8	C(Education)[T.2]	2.261598
21	C(JobRole)[T.Research Director]	2.231553
28	Age	2.104707
17	C(JobRole)[T.Human Resources]	2.049146
20	C(JobRole)[T.Manufacturing Director]	1.940330
43	YearsSinceLastPromotion	1.719382
26	C(MaritalStatus)[T.Single]	1.711571
25	C(MaritalStatus)[T.Married]	1.690324
11	C(Education)[T.5]	1.303642
35	NumCompaniesWorked	1.291810
39	TrainingTimesLastYear	1.040054
27	C(OverTime)[T.Yes]	1.036543
12	C(Gender)[T.Male]	1.033266
30	EnvironmentSatisfaction	1.033092

32	JobInvolvement	1.029939
40	WorkLifeBalance	1.028596
29	DistanceFromHome	1.028264
37	RelationshipSatisfaction	1.028146
34	MonthlyRate	1.027966
31	HourlyRate	1.025140
33	JobSatisfaction	1.024686
36	PercentSalaryHike	1.024530

Given the high P values, we should consider removing these from our model

Variable	p-value
Education 2-5	0.60-0.94
HourlyRate	0.656
PercentSalaryHike	0.584
YearsAtCompany	0.704

Then these next | Variable | p-value | | ----- | ----: | | JobLevel 3 | 0.936 | | JobLevel 4 | 0.528 | | JobLevel 5 | 0.466 | | JobRole Manager | 0.846 | | JobRole Research Scientist | 0.834 | | Manufacturing Director | 0.441 |

Fit

Final Model

The final model included:

- Param 1
- Param 2

```
# fit_log.summary()
```

The fitted model produced:

- $R^2 = 0.818$
- Adjusted $R^2 = 0.812$
- F-statistic = 141.6
- $p < 0.001$

indicating strong overall explanatory power.

Important Continuous Predictors

Variable	Coefficient
OverTime	2.07

Variable	Coefficient
Business Travel	2.02/1.05
Marital Status	1.60

OverTime

Employees working overtime have roughly 8 times the odds of attrition compared with employees who do not, holding the other variables constant:

$$e^{2.07} \approx 7.95$$

Business Travel

Employees traveling frequently have roughly 7.5 times the odds of attrition compared to employees who do not travel, holding the other variables constant:

$$e^{2.02} \approx 7.5$$

Employees traveling rarely also have roughly 2.9 times the odds of attrition compared to employees who do not travel, holding the other variables constant:

$$e^{1.05} \approx 2.9$$

Marital Status

Single employees have roughly 5 times the odds of attrition compared to divorced employees:

$$e^{1.60} \approx 5$$

Evaluate

Residual Analysis

Residual diagnostics indicated:

- No severe violations of linearity.
- Acceptable residual symmetry.
- Some heavy tails remain.
- No extreme multicollinearity concerns.

The largest VIF values were associated with popular truck categories but remained below commonly cited thresholds requiring corrective action.

Limitations

Several limitations should be noted:

- No MSRP information was available.
- Trim level information was not incorporated.
- Accident history was unavailable.
- Maintenance records were unavailable.
- Auction data may not represent all vehicle markets.

These limitations likely explain a portion of the unexplained variance.

Predict

Based off of the given results, we do see that the model does give more value to vehicles in better condition and those that are newer. I'm slightly surprised that it gives more value to certain models over others but it does make sense considering that certain models are more sought after than others. It does indicate that trucks retain more value over sedans. These predictions seem reasonable and can be used to inform purchasing decisions.

Business Implications

The model suggests:

1. Vehicle age is the strongest continuous driver of depreciation.
2. Mileage has a significant but smaller effect.
3. Vehicle model contributes substantial value differences.
4. Trucks generally retain value better than passenger sedans.

Recommendations

Organizations purchasing used vehicles should:

- Prioritize lower-age vehicles when possible.
- Consider mileage alongside age rather than in isolation.
- Use model-specific adjustments when evaluating value.
- Compare observed asking prices against model predictions to identify unusually favorable purchasing opportunities.

Future Work

Future improvements could include:

1. Incorporating original MSRP.
2. Including trim-level information.
3. Adding accident-history variables.
4. Incorporating local economic indicators.

Conclusion

The final regression model explains approximately 82% of the variation in used vehicle selling prices and provides a practical framework for vehicle valuation. Vehicle age, mileage, model, and geographic location all contribute significantly to price determination. The resulting model can support data-driven purchasing and pricing decisions while highlighting areas for future model enhancement.